

LAC LA BICHE, AB, Canada

WMO: 713160

Lat: 54.7689N

Lon: 112.0236W

Elev: 1860

StdP: 13.73

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.4	-22.4	-36.5	0.7	-27.3	-30.7	1.0	-21.6	20.5	32.4	18.1	28.6	2.3	170	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	81.5	63.1	78.3	61.4	75.4	60.1	65.9	77.3	63.8	74.5	61.9	71.8	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	87.8	70.9	59.6	81.5	68.4	57.8	76.4	66.3	31.7	77.5	30.0	74.8	28.6	71.8	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	15.8	13.9	-37.3	87.0	6.8	3.2	-42.2	89.3	-46.2	91.1	-50.0	92.9	-54.9	95.2		
			-36.9	69.4	6.3	2.2	-41.4	71.0	-45.1	72.3	-48.6	73.5	-53.2	75.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	6.9	11.6	22.7	37.9	50.5	58.2	62.9	60.3	51.4	38.5	22.0	9.4
	DBStd	22.80	16.98	14.43	14.28	9.95	8.28	5.66	4.97	5.78	7.60	8.23	12.06	15.13
	HDD50	6271	1335	1075	846	374	95	7	0	3	74	363	841	1258
	HDD65	10594	1800	1495	1311	812	453	213	99	166	410	821	1291	1723
	CDD50	1219	0	0	0	12	110	253	401	321	115	7	0	0
	CDD65	67	0	0	0	0	3	9	35	19	1	0	0	0
	CDH74	924	0	0	0	3	75	160	376	255	54	1	0	0
	CDH80	155	0	0	0	0	13	23	72	39	8	0	0	0
Wind	WSAvg	6.0	5.3	5.6	6.3	6.8	6.8	6.1	5.9	5.6	6.2	6.6	5.9	5.4
Precipitation	PrecAvg	15.9	0.7	0.5	0.6	0.9	1.6	2.8	3.1	2.0	1.3	0.7	0.6	0.7
	PrecMax	20.0	1.5	1.1	1.3	2.2	3.4	5.4	6.1	3.5	2.9	2.2	1.6	1.5
	PrecMin	11.8	0.0	0.0	0.1	0.1	0.2	0.9	0.9	0.8	0.4	0.2	0.2	0.0
	PrecStd	2.4	0.4	0.3	0.4	0.5	0.7	1.2	1.1	0.7	0.7	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	44.7	53.3	72.0	81.3	82.9	86.1	84.2	80.1	68.0	50.9	45.3
		MCWB	37.1	36.3	41.1	51.4	57.8	61.6	66.7	65.8	60.9	52.2	41.5	36.7
	2%	DB	39.9	39.8	47.5	64.4	75.2	78.4	81.6	80.1	74.0	59.7	44.5	39.6
		MCWB	34.4	33.6	37.7	47.4	55.5	59.3	64.4	64.0	57.9	47.4	38.0	34.1
	5%	DB	35.7	35.6	44.1	59.0	70.8	74.8	78.1	76.6	69.5	55.3	40.5	34.7
		MCWB	31.7	30.8	35.8	44.3	52.5	57.9	62.9	62.0	55.9	45.0	35.2	30.5
	10%	DB	30.9	31.9	40.5	53.9	66.7	71.2	75.0	73.3	65.0	51.4	36.8	30.0
		MCWB	27.6	28.3	33.8	41.8	50.5	56.5	61.4	60.4	53.8	42.8	32.6	27.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	37.3	41.9	51.7	59.8	64.7	69.1	68.4	62.2	52.6	42.3	37.7
		MCDB	43.6	42.7	51.7	70.0	75.2	77.8	80.8	80.0	76.1	65.9	48.5	44.0
	2%	WB	34.9	34.0	38.5	48.0	56.7	61.9	66.7	65.5	59.2	48.6	38.8	34.2
		MCDB	39.4	39.2	46.3	62.6	71.5	73.0	78.0	76.8	71.5	57.9	43.7	39.4
	5%	WB	31.9	31.1	36.3	45.0	54.3	59.9	64.6	63.3	57.0	45.9	35.7	30.7
		MCDB	35.9	35.4	43.1	57.8	67.7	70.9	75.5	73.5	67.8	53.8	39.9	34.3
	10%	WB	27.6	28.1	34.1	42.3	51.9	58.2	62.7	61.4	54.8	43.5	32.9	27.2
		MCDB	30.7	31.9	39.9	52.8	64.2	68.4	72.3	70.9	63.3	50.4	36.3	30.0
Mean Daily Temperature Range	5% DB	MDBR	16.6	19.1	18.8	20.2	22.3	20.4	19.7	20.4	20.1	15.9	13.8	15.7
		MCDBR	19.9	22.4	21.2	28.3	29.3	27.3	25.4	26.2	28.4	24.3	18.0	19.7
		MCWBR	15.8	17.3	13.4	15.7	13.4	12.0	11.6	12.3	14.5	14.2	12.5	15.6
	5% WB	MCDBR	19.5	21.5	20.2	27.1	26.4	22.9	23.2	24.2	26.4	23.0	16.8	19.5
		MCWBR	15.6	17.1	13.2	15.5	12.8	11.3	11.7	12.1	13.9	13.9	12.1	15.8
Clear-Sky Solar Irradiance	taub	0.251	0.266	0.293	0.332	0.345	0.361	0.360	0.367	0.316	0.284	0.255	0.237	
	taud	2.295	2.325	2.369	2.388	2.403	2.367	2.430	2.352	2.511	2.576	2.457	2.287	
	Ebn at Noon	226	265	282	284	285	281	279	269	272	260	229	206	
	Edh at Noon	17	24	29	33	34	36	33	34	25	19	15	14	
All-Sky Solar Radiation	RadAvg	258	580	1060	1467	1785	1839	1853	1519	1057	583	268	181	
	RadStd	27	44	79	119	161	127	101	115	110	89	32	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page